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(54) Title of the invention : A device for segregating waste and a method of working thereof

(51) International classification	:B65F 1/14, B65F 1/00, B07C 5/36, B65F 1/16, B07C 5/342	(71)Name of Applicant : 1)BGS College of Engineering and Technology (BGSCET) Address of Applicant :adjacent to Mahalakshmi Metro Station, Mahalakshmiapuram, West of Chord Road, Bengaluru – 560 086, Karnataka, India Bangalore Karnataka India
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(57) Abstract :

The present invention provides a waste-segregation device (100) that addresses inaccurate manual sorting and inefficient multi-bin usage. The solution comprises a structural frame (200) defining bin compartments containing multiple waste bins (300). An upper platform (400) receives deposited waste items, while a user interface display (500) mounted on the support bar (200g) includes a camera (500a) for capturing waste images and a touch display (500b) for user interaction. A processing unit (600) classifies the captured images using a convolutional neural network and determines the appropriate bin (300a to 300f). Each bin is positioned above a load cell (300aa to 300dd) for weight measurement and includes an ultrasonic sensor (300aaa to 300ddd) for fill-level detection. A pan-tilt sorting mechanism associated with the platform (400a) employs a movable chute and servo motors to direct the item into the selected compartment. Operational data may be transmitted to a remote server for system optimisation.

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